

MCQs

Set 'A'

Group 'A'

1. (d) 400 ml
2. (b) 25 ml OF a 0.1M solution
3. (a) Mol. wt.
4. (d) Na_2SO_4
5. (b) 0.01M CaCl_2
6. (b) $\text{C}_4\text{H}_2\text{O}$
7. (b) Smelting with Carbon
8. (c) wrought iron
9. (c) Methanoic acid
10. (d) SOCl_2
11. (d) Benzoyl Chloride

MCQs

Group 'A'

Set 'B'

1. (b) H_2O
2. (c) wrought iron
3. (c) Methanoic acid
4. (d) Na_2SO_4
5. (d) Benzoyl Chloride
6. (d) 400 ml
7. (b) 25 ml OF a 0.1M solution
8. (a) Mol. wt.
9. (d) SOCl_2
10. (b) smelting with Carbon
11. (b) 0.01M CaCl_2

Group 'B'

Q.N. 1. (a) Ans: we have,

$$\text{Normality (N)} = \frac{\text{no. of gram equivalent}}{\text{Volume of solution in ml}} \times 1000 \dots (1)$$

Page _____

$$\text{no. of gram equivalent} = \frac{\text{Total weight}}{\text{equivalent weight}} = \frac{w}{E} \dots (2)$$

from eqn (1) & (2),

$$\text{Normality (N)} = \frac{w}{E} \times \frac{1}{\text{volume of solution in ml}} \times 1000$$

$$\text{or, } N = \frac{w}{E} \times \frac{1}{V_{ml}} \times 1000$$

$$\text{or, } \frac{w}{E} = \frac{N V_{ml}}{1000}$$

For acid,

$$\text{no. of gram equivalent} = \left(\frac{N V_{ml}}{1000} \right) \text{ acid}$$

For base,

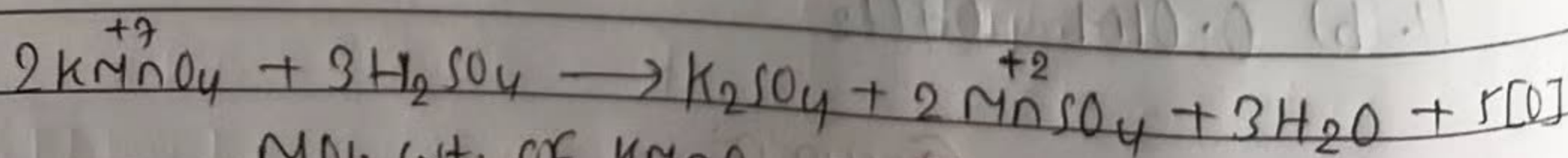
$$\text{no. of gram equivalent} = \left(\frac{N V_{ml}}{1000} \right) \text{ base}$$

for neutralization,

$$\text{or, } \left(\frac{N V_{ml}}{1000} \right) \text{ acid} = \left(\frac{N V_{ml}}{1000} \right) \text{ base}$$

$$\therefore N_a V_a = N_b V_b$$

1.6) Ans: In acidic medium,



Mol. wt. of $\text{KMnO}_4 = 158$

Change in O.N. of Mn = $7 - 2 = 5$

$$\therefore \text{Equivalent wt} = \frac{158}{5} = 31.6 //$$

2)

1. (C) Ans: Titration is used in pharmaceutical industry in:

- To maintain content uniformity
- Stability studies
- Dissolution Testing

Q. N. 2. (a) Ans: The differences between titrant and titrand:

Titrant	Titrand
1. It's concentration is known.	1. It's concentration is unknown.
2. It's normally kept in burette.	2. It's normally kept in conical flask.
3. It's volume is measured.	3. It's volume is already known.
4. Generally, It's Primary standard solution.	4. Generally, It's Secondary standard solution.

2. (b) Solution:

For HCl,

$$M_1 = 0.1M$$

$$N_1 = 0.1N$$

$$V_1 = ?$$

For Na_2CO_3 and $NaHCO_3$,

$$\text{wt. of mixture} = 1 \text{ gm}$$

$$\text{wt. of } Na_2CO_3 (w_1) = x \text{ (let)}$$

$$\text{wt. of } NaHCO_3 (w_2) = 1 - x$$

we have,

$$(\text{No. of moles})_{Na_2CO_3} = (\text{No. of moles})_{NaHCO_3}$$

$$\text{or, } \frac{x}{106} = \frac{1-x}{84}$$

$$\text{or, } 84x = 106 - 106x$$

$$\therefore x = 0.557 \text{ gm}$$

wt. of Na_2CO_3

$$\text{Wt. of NaHCO}_3 = 1 - x = 1 - 0.557 = 0.442 \text{ gm}$$

Now,

$$\frac{N_1 V_1}{1000} = \frac{W_1}{E_1} + \frac{W_2}{E_2}$$

$$\text{Or, } \frac{0.1 \times V_1}{1000} = \frac{0.557}{53} + \frac{0.442}{84}$$

$$\therefore V_1 = 157.71 \text{ ml} //$$

Hence, volume of HCl required is 157.71 ml.

Q.N.3. (a) Ans: The difference between solubility product and ionic product:

Solubility Product	Ionic Product
1. It is the ionic product of saturated solution at given temperature.	1. It is the molar concentration of any type of solution at given temperature.
2. It is only defined for a saturated solution.	2. It is defined for any kinds of solution.
3. Precipitation does not occur if $K_{sp} > K_{ip}$.	3. Precipitation occurs if $K_{sp} > K_{ip}$.
4. It is denoted by K_{sp} .	4. It is denoted by K_{ip} .

3.(b) Solution:

For acid,

$$pH = 1$$

$$[H^+] = 10^{-1} M = N_1$$

for base,

$$pH = 12$$

$$pOH = 14 - pH = 14 - 12 = 2$$

$$[H^+] = 10^{-2} M = N_2$$

$$V_1 = x$$

$$V_2 = x$$

$$V_{mix} = 2x$$

Now,

$$N_{mix} \times V_{mix} = N_1 V_1 + N_2 V_2$$

$$\text{or, } N_{mix} \times 2x = 10^{-1} \times x + 10^{-2} \times x$$

$$\text{or, } N_{mix} \times 2x = x (10^{-1} + 10^{-2})$$

$$\therefore N_{mix} = \frac{x (10^{-1} + 10^{-2})}{2x} = 0.045 \text{ m}$$

$$\text{Also, } \text{pH} = -\log [H^+] = -\log [0.045] = 1.346$$

Hence, pH of resulting solution is 1.346. //

Q. N. 4. (a) Ans:

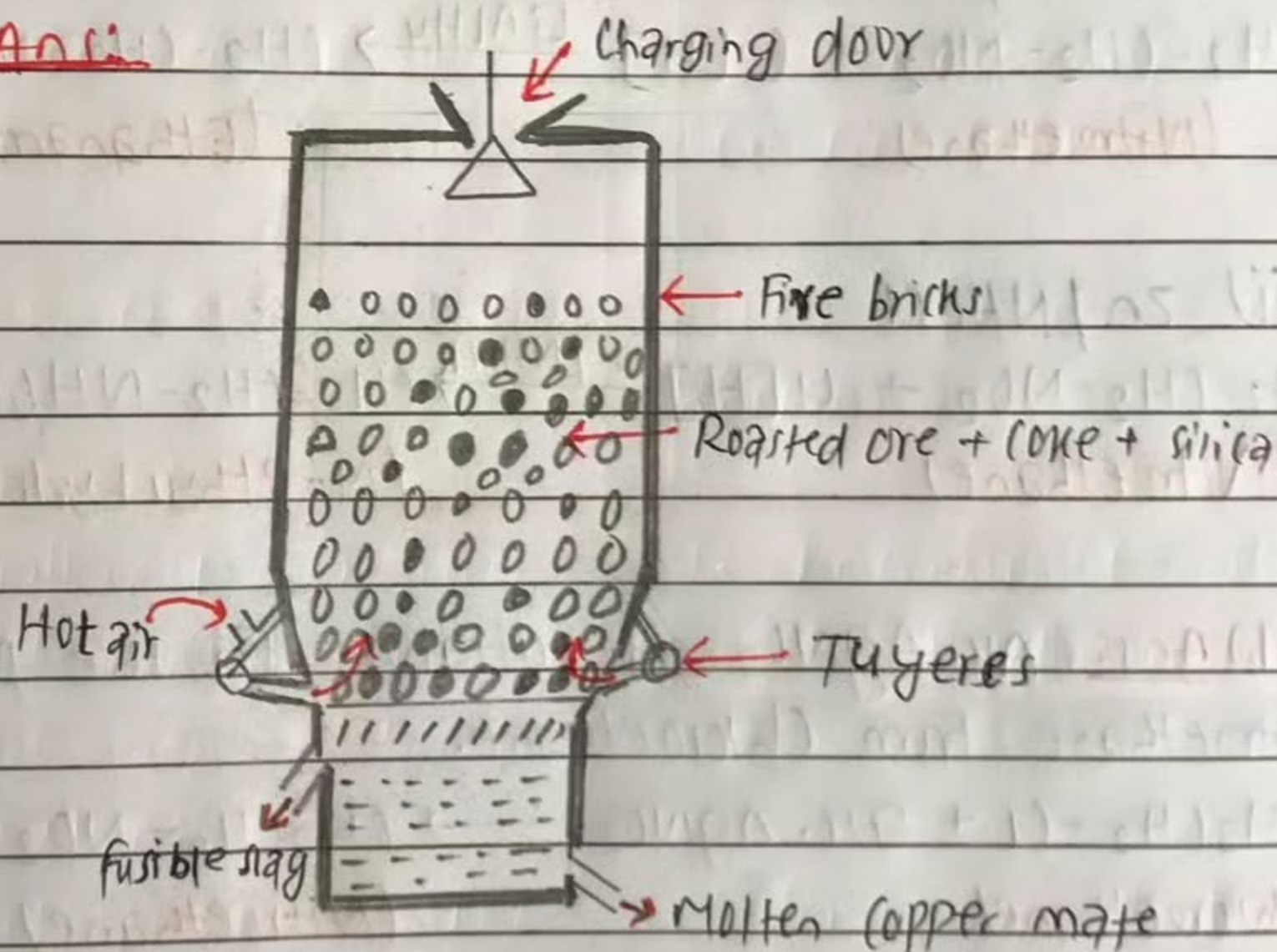
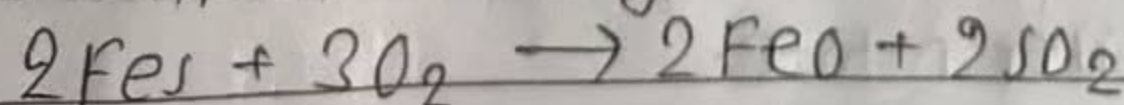


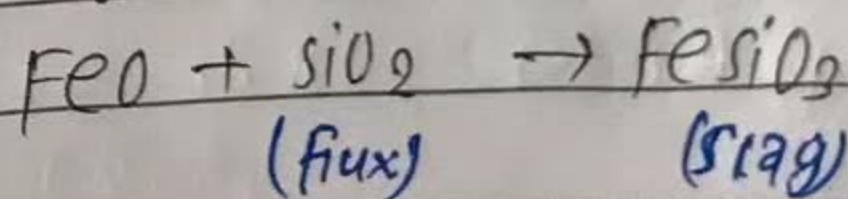
Fig: Blast furnace

Q. (b) Ans: The different chemical reactions involved during the formation of matte in the furnace:

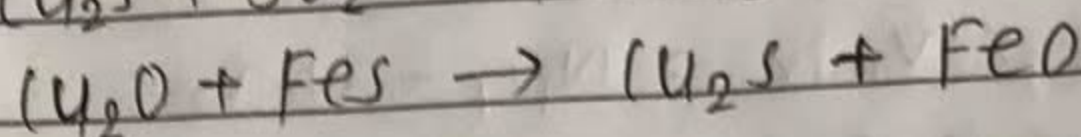
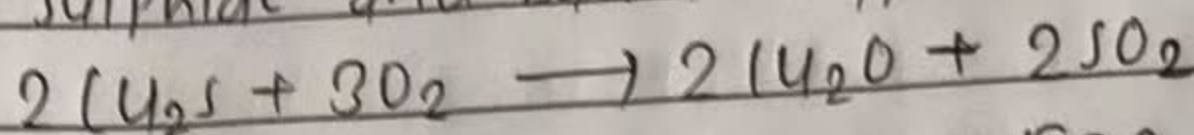
(i) Iron sulphide changes into ferrous oxide.



(ii) Ferrous oxide combines with silica to form ferrous silicate.

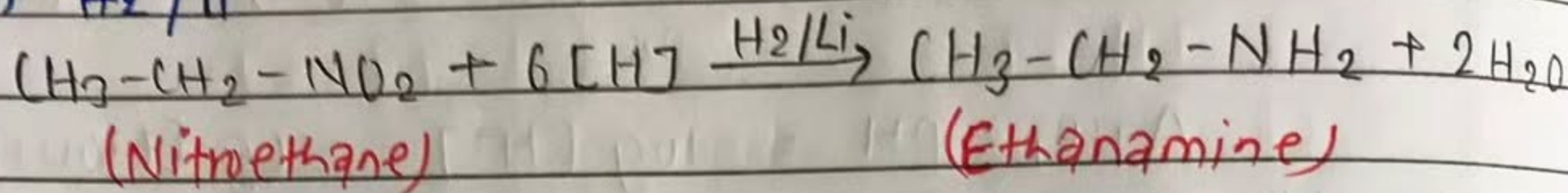


(iii) Copper oxides formed during smelting react with ferrous sulphide and back to cuprous sulphide.

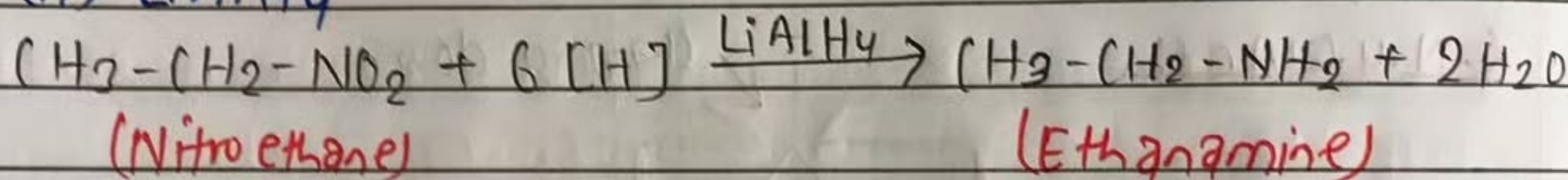


Q.N.5. (a) Ans:

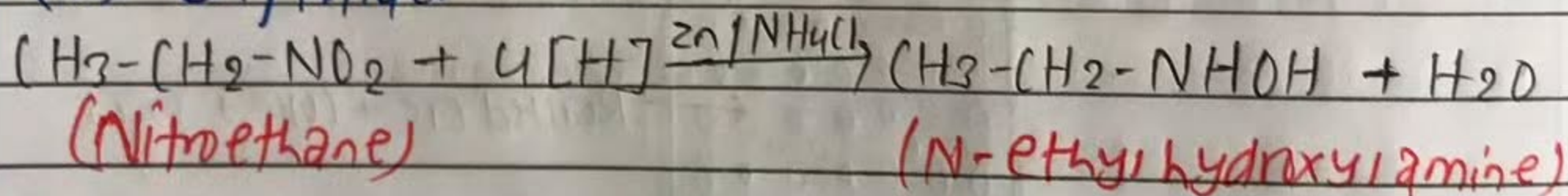
(i) H_2/Li



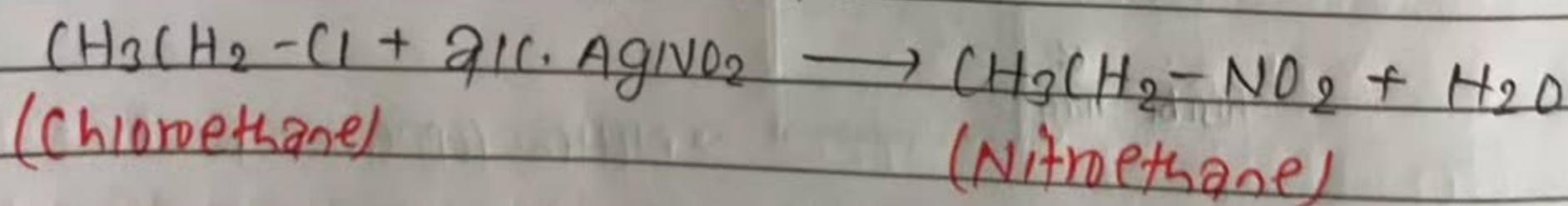
(ii) LiAlH_4



(iii) $\text{Zn}/\text{NH}_4\text{Cl}$



5.(b) Ans: A/c. AgNO_2 is suitable for the preparation of nitroethane from chloroethane.

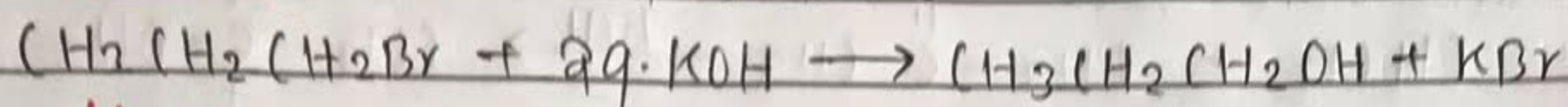


In a/c. KNO_2 , NO_2^- acts as nucleophile to attack either through N-atom or O-atom but C-O bond is stronger than C-N bond thus formed ethanenitrite.

But in a/c. AgNO_2 , it gives nitroethane due to covalent nature of silver nitrite.

P.T.O. $\rightarrow$

Q.N.6(a) Ans: 'B' $\rightarrow$ Propan-1-ol

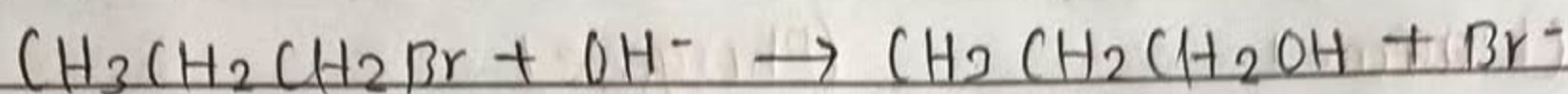


(1-Bromopropane)

(Propan-1-ol)

(B)

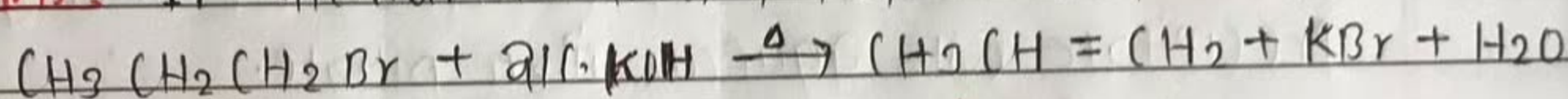
6.(b) Ans: Nucleophilic substitution reaction $\text{S}_\text{N}2$ mechanism occurs in the above reaction because 1-Bromopropane being a primary alkyl halide, which favors backside attack by the nucleophile. OH^- acts as strong nucleophile than Br^- .



(1-Bromopropane)

(Propan-1-ol)

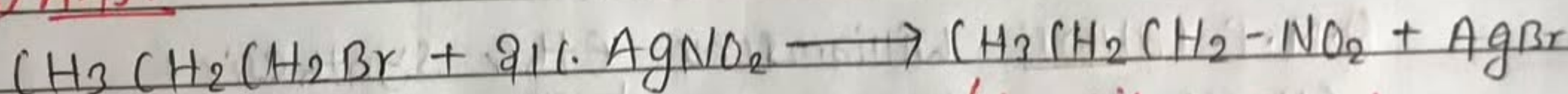
6.(c) Ans: If alc. KOH is used, Propene is formed.



(1-Bromopropane)

(propene)

6.(d) Ans:



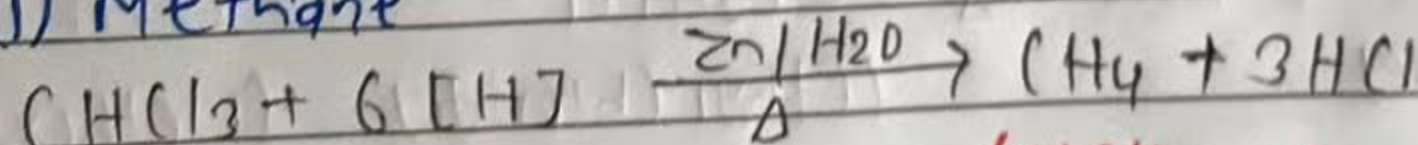
(1-Bromopropane)

(1-Nitropropane)

(A)

Q.N.7 (a) Ans:

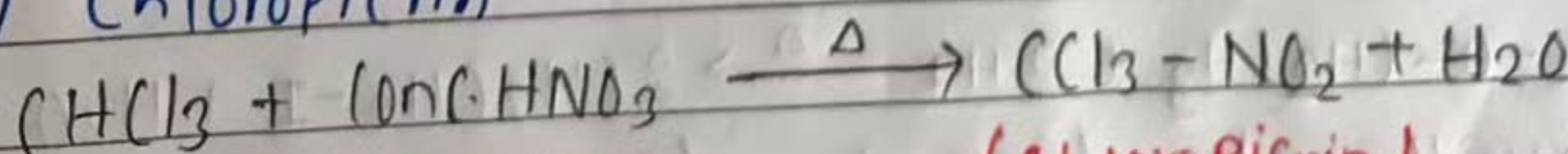
(i) Methane



(Trichloromethane)

(Methane)

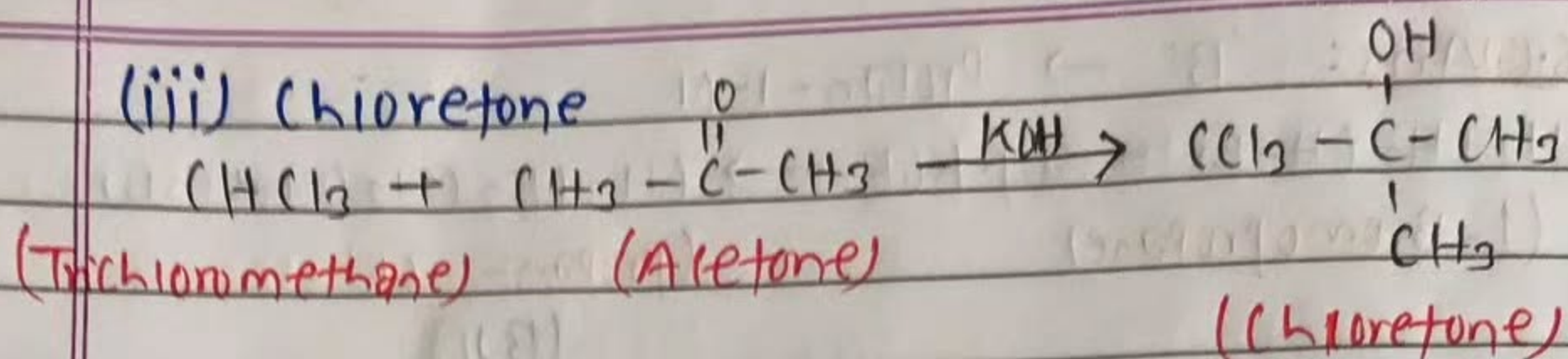
(ii) Chloropicrin



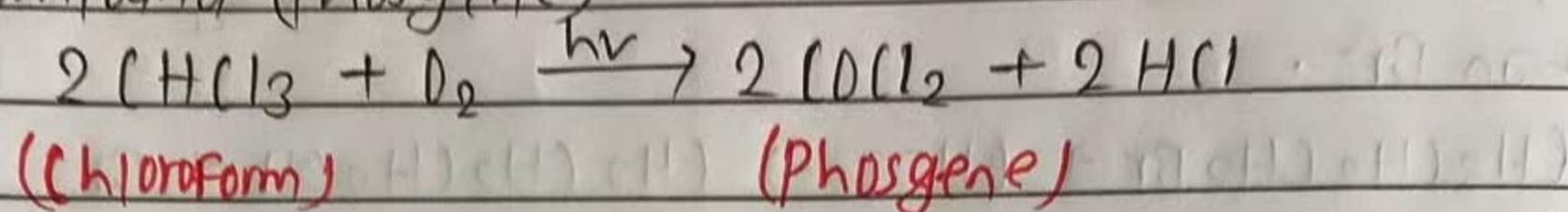
(Trichloromethane)

(Chloropicrin)

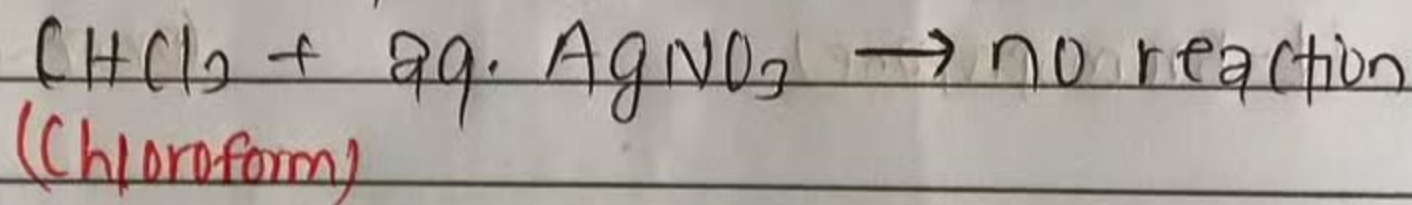
(Tear gas)



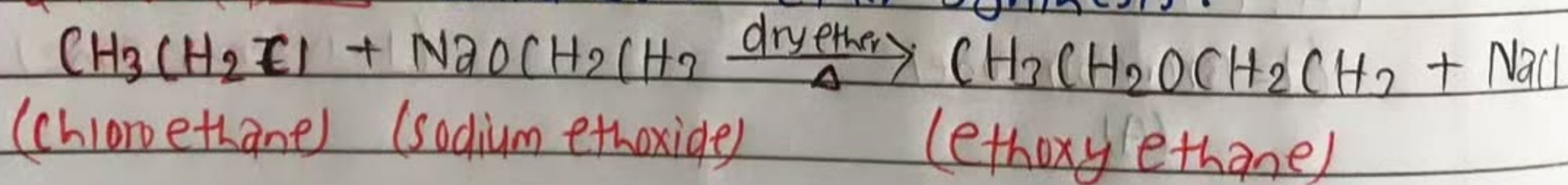
7.(b) Ans: Dark brown airtight bottle (Kts off sunlight and prevent the oxidation of Chloroform into poisonous compound (Phosgene).



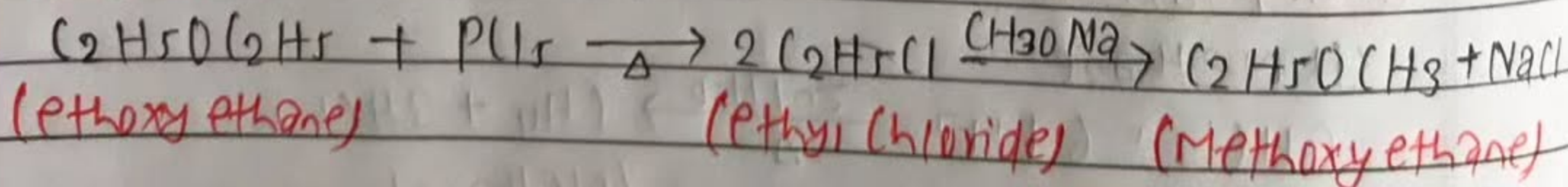
7.(c) Ans: Chloroform is covalent compound and doesn't contain free Cl^- ions. So, Chloroform does not give white precipitate with aqueous silver nitrate.



Q.N.P. (a) Ans: Williamson's ether synthesis:

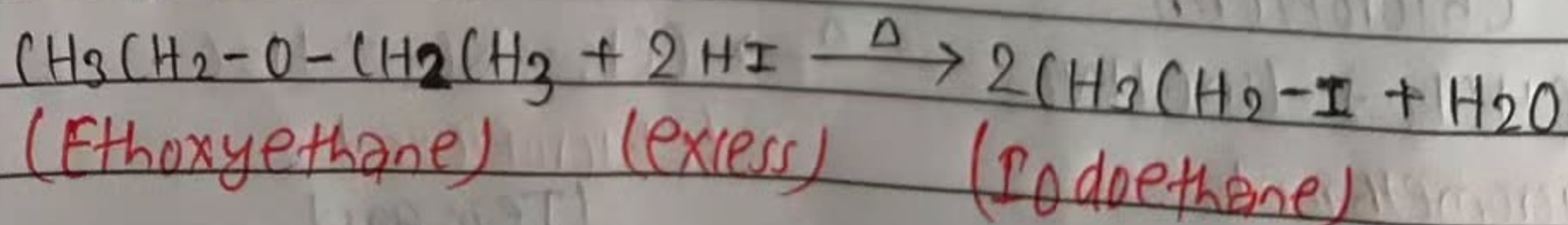


8.(b) Ans:



8.(c) Ans:

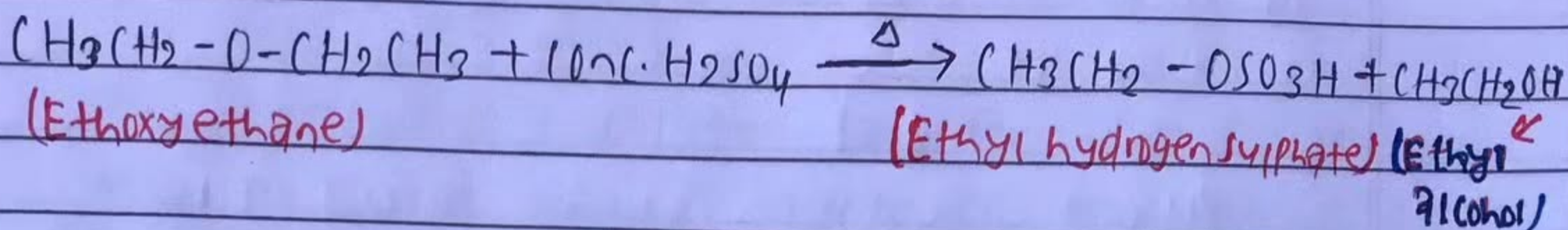
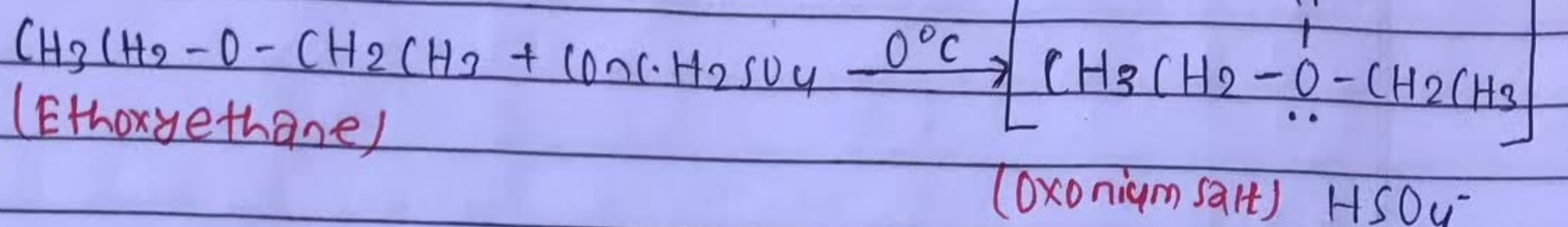
(i) Excess HI



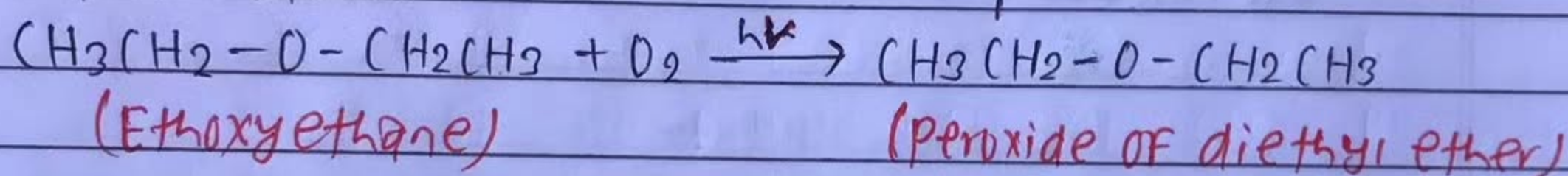
P.T.O. $\rightarrow$

8.(c) Ans:

(ii) Conc. H_2SO_4



8.(d) Ans: Old sample of ether are not distilled to dryness in air because peroxide is formed which is highly unstable and explosive.



Group 'C'

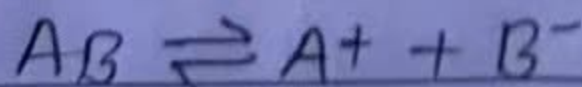
Q.N.9. (a) Ans: Let us consider an electrolyte AB which ionizes as: $AB \rightleftharpoons A^+ + B^-$

Applying law of mass action,

$$K = \frac{[A^+][B^-]}{[AB]} \dots (1)$$

where, K is ionization or dissociation constant.

If 'c' be initial concentration of electrolyte and 'x' be degree of dissociation.



$$\text{At } t = 0 \quad c \quad 0 \quad 0$$

$$\text{At equilibrium } c(1-x) \quad cx \quad cx$$

Substituting these values in eqn (1),

$$K = \frac{cx \cdot cx}{c(1-x)} = \frac{x^2 c}{1-x} \dots (2)$$

For weak electrolyte, $\alpha \ll 1$. So, $1 - \alpha \approx 1$.

From eqn (2),

$$K = \alpha^2 C$$

$$\text{Or, } \alpha^2 = \frac{K}{C}$$

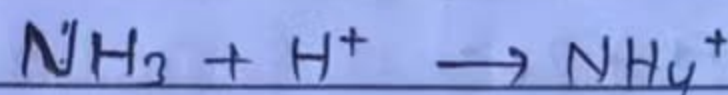
$$\therefore \alpha = \sqrt{\frac{K}{C}}$$

This is Ostwald's dilution law for weak electrolytes.

Also, it states that, "degree of ionisation of weak electrolyte is increased with dilution at constant temperature."

$$\text{i.e. } \alpha \propto \sqrt{K_c \times V}$$

9.(b) Ans:



Here, NH_3 accepts H^+ ion to form NH_4^+ . So, NH_3 is Brønsted base and NH_3 does not give OH^- in water. So, NH_3 is not Arrhenius base.

9.(c) Ans: $\text{CuSO}_4 + \text{H}_2\text{O} \rightarrow \text{Cu(OH)}_2 + \text{H}_2\text{SO}_4$

(weak base) (strong acid)

Here, No. of H^+ is more than OH^- ions. Hence, aqueous solution of CuSO_4 is acidic in nature.

9.(d) Ans: The chemical substance which indicates the end point of titration is called indicator.

For strong base and weak acid,

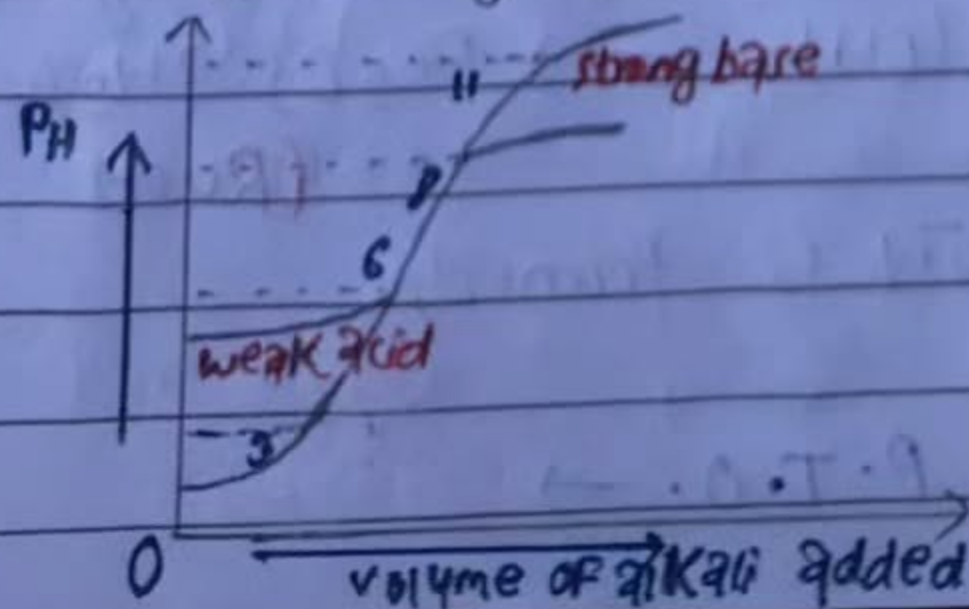


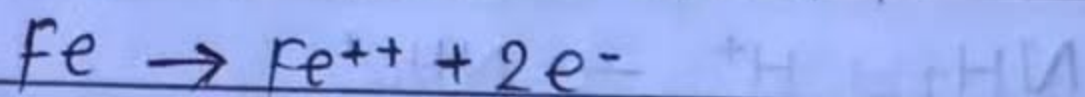
Fig:- Titration curve

Here, pH rapidly varies from 6-11. pH of Phenolphthalein (8.2-10) lies in this range. Hence, It is a suitable indicator for the titration.

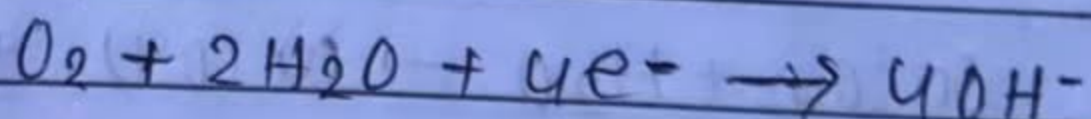
Q.N. 10. (a) Ans: Rusting is the formation of brown layer of $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ in the surface of iron when exposed in moist air for a long time.

According to electrochemical theory of rusting, the surrounding acts as voltmeter cell for the reaction of electrolysis of iron.

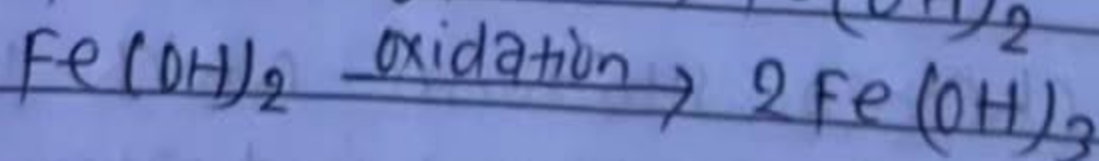
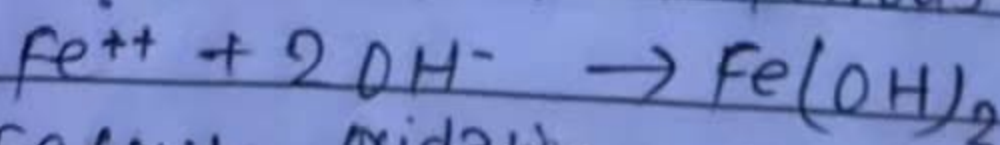
At Anode: When Fe is placed in surrounding, it loses its two electrons to form Fe^{2+} .



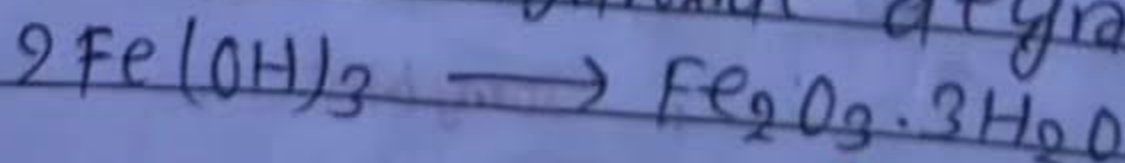
At Cathode: Electron liberated at anode move to region of higher electric potential where reduction of oxygen into OH^- ions.



Ferrous ion produced at anode and OH^- ion at cathode combine to form ferrous hydroxide.



Thus, formed ferrous hydroxide degrades giving rust.



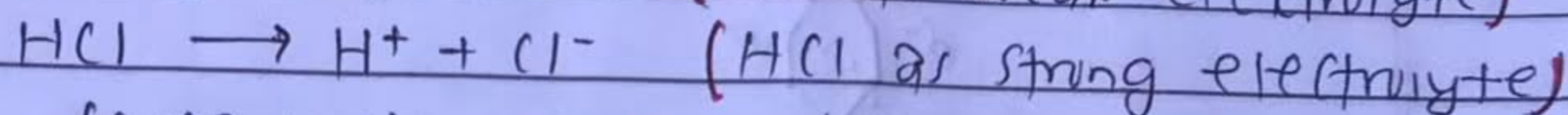
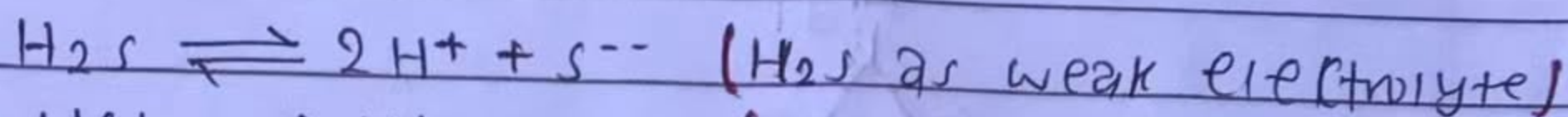
(Rust)

Hence, rust is formed.

P.T.O. $\rightarrow$

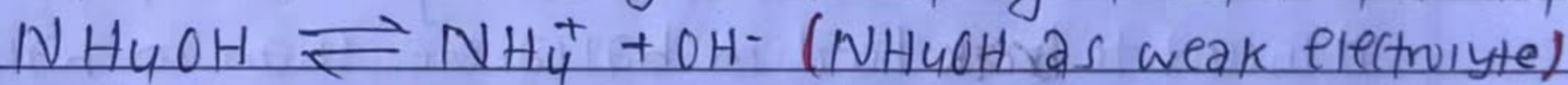
10.(b) Ans: The application of common ion effect and solubility product principle in qualitative salt analysis:

(i) **precipitation of Group II metal ions:** They are precipitated as metal sulphides by passing H_2S gas to HCl solution of salt.



Here, concentration of S^{--} decreases due to common ion effect. Only solubility product of Group II metal sulphides are exceeded by their ionic product.

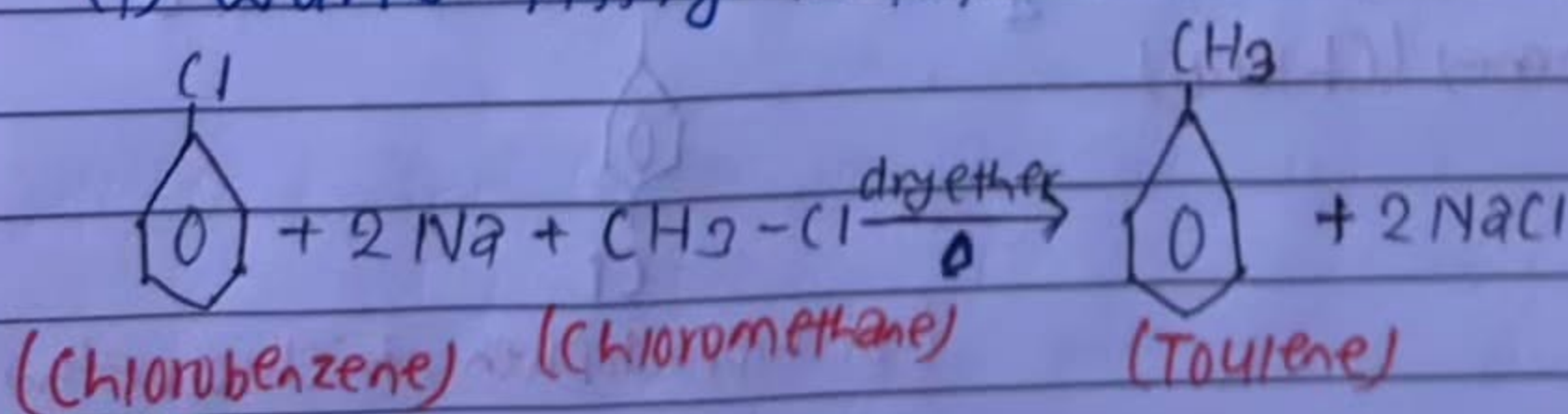
(ii) **precipitation of Group IIIA metal ions:** They are precipitated as metal hydroxide using NH_4OH and NH_4Cl .



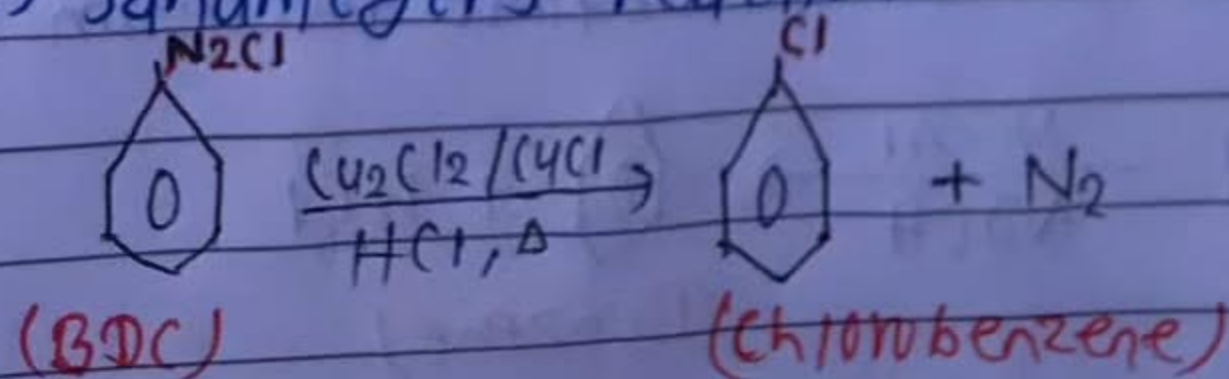
Here, concentration of OH^- decreases due to common ion effect. Only solubility product of Group IIIA metal hydroxide are exceeded by their ionic product.

Q. N. 11. (2) Ans:

(i) **Wurtz-Fittig reaction**



(ii) **Sandmeyer's reaction**



11. (b) Ans:

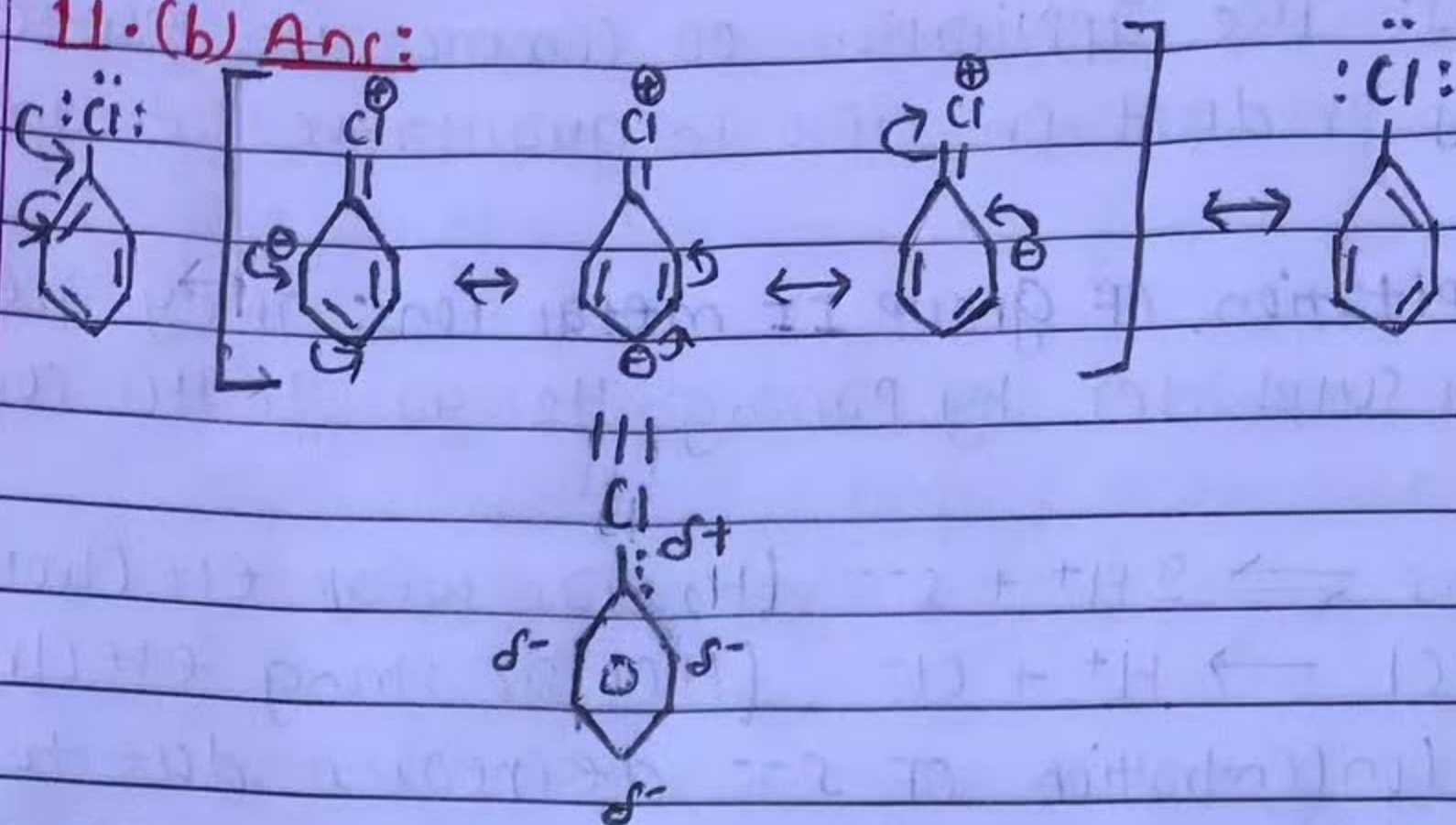
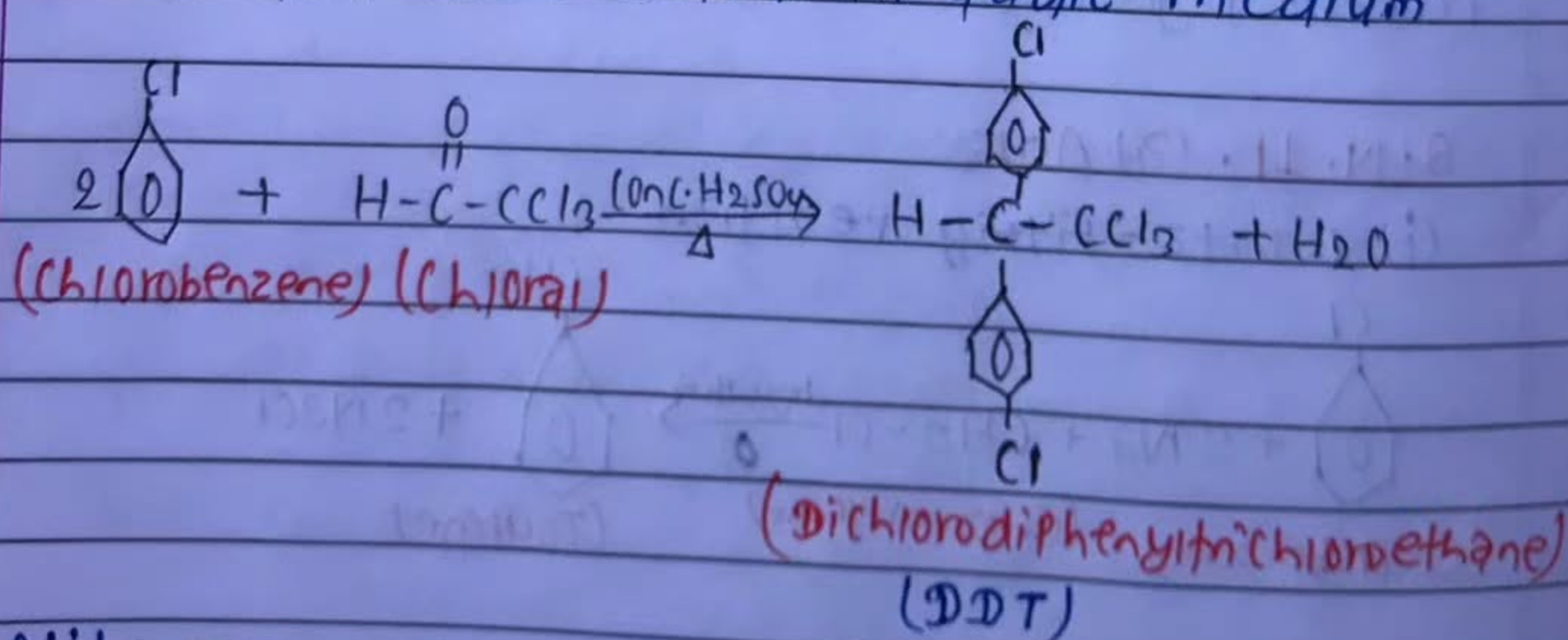


Fig: Resonating structure of Chlorobenzene

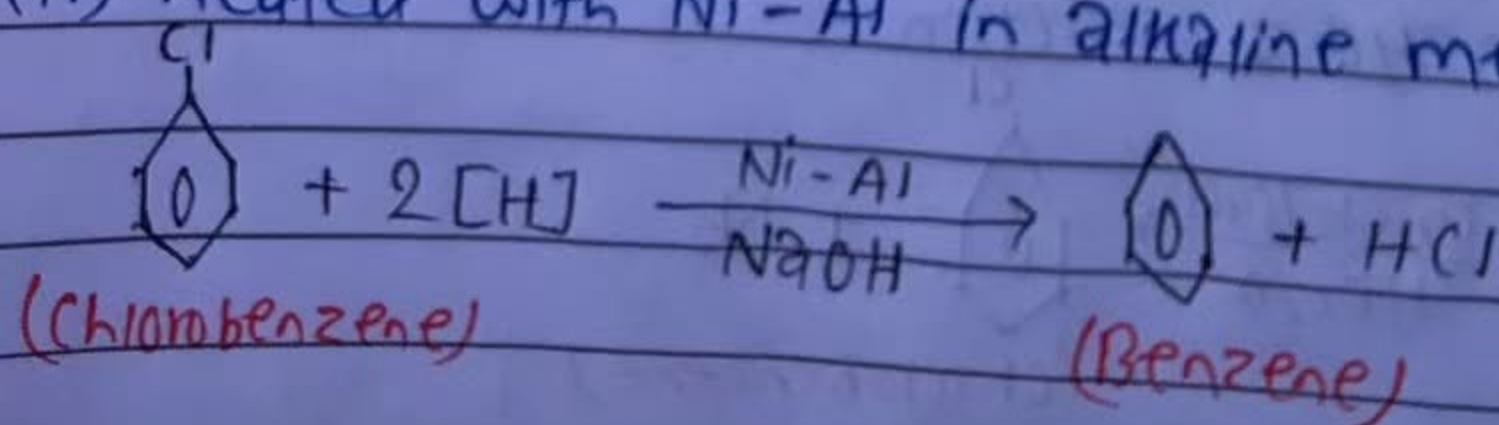
From above resonating structure, It is clear that C-X bond in haloarene has partial double bond character (sp^2 hybridized C-atom) which is shorter than haloalkane. C-X bond is less polar and stronger. Thus, It is more difficult to ~~break~~ bond in haloarene than in haloalkane due to electron withdrawing Phenyl group.

11. (c) Ans:

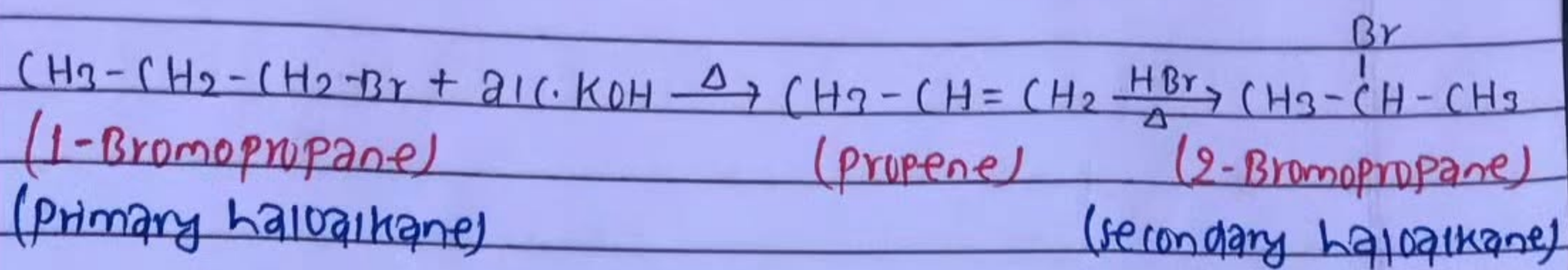
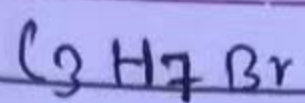
(i) treated with Chloral in acidic medium



(ii) heated with Ni-Al in alkaline medium



11. (d) Ans:



"The END"

FIRST TERM EXAMINATION 2082

SET- 'A'

Class: XII 'Science'

Time: 25 Min.

Subject: Chemistry (3021)

Tick the best options among the four options. Apply tick on two options will be automatically denied and mark will not be counted.

GROUP - 'A' (MCQ) [11×1 = 11]

- Volume of water be added to $N/2$ HCl to prepare 500 cm³ of $N/10$ solution is
a) 100 ml b) 200 ml c) 300 ml d) 400 ml
- Which of the following solutions of sulphuric acid will exactly neutralize 25 ml of 0.2 M sodium hydroxide solution?
a) 12.5 ml of a 0.1 M solution b) 25 ml of a 0.1 M solution
c) 25 ml of a 0.2 M solution d) 50 ml of a 0.2 M solution
- Equivalent weight of Mohr's salt if it acts as reducing agent is
a) Mol.wt. b) Mol.wt./3 c) Mol.wt./5 d) Mol.wt./6
- Which of the following salts does not undergo hydrolysis?
a) $CuSO_4$ b) $FeCl_3$ c) NH_4Cl d) Na_2SO_4
- Solubility of AgCl will be minimum in
a) 0.01 M NaCl b) 0.01 M $CaCl_2$
c) pure water d) 0.001 M $AgNO_3$
- Extremely hot copper wire reacts with steam to give
a) CuO b) Cu_2O c) Cu_2O_2 d) CuO_2
- Iron is extracted on large scale from Fe_2O_3 by
a) Reduction with Al b) Smelting with carbon
c) Calcination d) Roasting
- The purest form of iron is
a) White cast iron b) Grey cast iron
c) Wrought iron d) Steel

9. When chloroform is boiled with aqueous KOH followed by acidification, it gives

- a) Methoxy methane
- c) Methanoic acid

- b) Methanol
- d) Ethanol

10. The best reagent to convert alcohol into alkyl chloride is

- a) PCl_3
- c) $HCl/ZnCl_2$

- b) PCl_5
- d) $SOCl_2$

11. Chlorine gas reacts with benzaldehyde to give

- a) Chlorobenzene
- c) Benzal chloride

- b) Benzyl Chloride
- d) Benzoyl chloride

GROUP - 'A' (MCQ) [11×1 = 11]

1. Extremely hot copper wire reacts with steam to give
a) CuO b) Cu_2O c) Cu_2O_2 d) CuO_2
2. The purest form of iron is
a) White cast iron b) Grey cast iron
c) Wrought iron d) Steel
3. When chloroform is boiled with aqueous KOH followed by acidification, it gives
a) Methoxy methane b) Methanol
c) Methanoic acid d) Ethanol
4. Which of the following salts does not undergo hydrolysis?
a) CuSO_4 b) FeCl_3 c) NH_4Cl d) Na_2SO_4
5. Chlorine gas reacts with benzaldehyde to give
a) Chlorobenzene b) Benzyl Chloride
c) Benzal chloride d) Benzoyl chloride
6. Volume of water be added to $N/2 \text{ HCl}$ to prepare 500 cm^3 of $N/10$ solution is
a) 100 ml b) 200 ml c) 300 ml d) 400 ml
7. Which of the following solutions of sulphuric acid will exactly neutralize 25 ml of 0.2 M sodium hydroxide solution?
a) 12.5 ml of a 0.1 M solution b) 25 ml of a 0.1 M solution
c) 25 ml of a 0.2 M solution d) 50 ml of a 0.2 M solution
8. Equivalent weight of Mohr's salt if it acts as reducing agent is
a) Mol.wt. b) Mol.wt./3 c) Mol.wt./5 d) Mol.wt./6

9. The best reagent to convert alcohol into alkyl chloride is
- a) PCl_3
 - b) PCl_5
 - c) $HCl/ZnCl_2$
 - d) $SOCl_2$
10. Iron is extracted on large scale from Fe_2O_3 by
- a) Reduction with Al
 - b) Smelting with carbon
 - c) Calcination
 - d) Roasting
11. Solubility of $AgCl$ will be minimum in
- a) 0.01 M $NaCl$
 - b) 0.01 M $CaCl_2$
 - c) pure water
 - d) 0.001 M $AgNO_3$

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

GROUP - 'B' [8×5=40]

Short answer questions

1. (a) Deduce normality equation. [2]
b) Calculate the equivalent weight of $KMnO_4$ in acidic medium including reaction involved. [2]
c) How would you apply titration in pharmaceutical industry? [1]
2. (a) Distinguish between titrant and titrand. [2]
b) How many ml of a 0.1 M HCl are required to react completely with 1 gm mixture of Na_2CO_3 and $NaHCO_3$ containing equimolar amount of the two. [3]
3. (a) Differentiate between solubility product and ionic product. [2]
b) If equal volume of two solutions having $pH = 1$ and $pH = 12$ are mixed together, what will be the pH of the resulting solution? [3]
4. A coinage metal (M) of electronic configuration $[Ar]4s^13d^{10}$ belongs to group IB in the periodic table.
a) Draw a blast furnace for the smelting process during the extraction of (M) using its chief ore. [2]
b) Explain the different chemical reactions involved during the formation of matte in the furnace. [3]
5. (a) What happens when nitroethane is reduced by
i) H_2/Li ii) $LiAlH_4$ iii) Zn/NH_4Cl [1+1+1]
b) Among *alc.* KNO_2 and *alc.* $AgNO_2$, which reagent is suitable for the preparation of nitroethane from chloroethane? Explain with reaction. [2]
6. A chemical reaction occurs as follows:
 $CH_3CH_2CH_2Br + aq. KOH \longrightarrow B + KBr$
(A)
a) Identify the compound 'B' in the above reaction and write its IUPAC name. [3]
b) Which substitution reaction mechanism occurs in the above reaction and why? [2]

- c) What is the product formed when *alc. KOH* solution is used instead of *aq. KOH* in the above reaction? [1]
- d) How can you obtain nitropropane from compound 'A'? [1]
7. a) Starting from trichloromethane, how would you prepare
 i) Methane ☒ ii) Chloropicrin ☒ iii) Chloretone ☒ [3]
- b) Why is chloroform stored in a dark brown airtight bottle? [1]
- c) Why does chloroform not give white precipitate with aqueous silver nitrate? [1]
8. a) Give an example of Williamson's ether synthesis. [1]
- b) Convert ethoxyethane into methoxyethane. [1]
- c) What happens when ethoxyethane is heated with:
 i) excess HI ☒ ii) *conc. H₂SO₄* [2]
- d) Why are old sample of ether not distilled to dryness in air? [1]

GROUP - 'C' [3X8=24]

Long answer questions.

9. a) Derive expression of Ostwald's dilution law for weak electrolyte and state the law. [2+1]
- b) NH_3 acts as Bronsted base but not Arrhenius base, why? [1]
- c) Why is aqueous solution of $CuSO_4$ acidic in nature? [1]
- d) Define indicator. How will you select suitable indicator for the titration between weak acid and strong base? [1+2]
10. a) What is rusting? Describe the electrochemical theory of rusting of iron. [1+4]
- b) Write the application of common ion effect and solubility product principle in qualitative salt analysis. [3]
11. a) Write the chemical equations to illustrate
 i) Wurtz-Fittig reaction ☒ ii) Sandmeyer's reaction [1+1]
- b) Why are haloalkanes more reactive than haloarenes towards nucleophilic substitution reaction? [2]
- c) What happens when chlorobenzene is
 i) treated with chloral in acidic medium? ☒
 ii) heated with $Ni - Al$ in alkaline medium? ☒ [1+1]
- d) Give proper reaction sequence to convert primary haloalkane into secondary haloalkane for the organic compound having molecular formula C_3H_7Br . [2]